

LangSmith vs Langfuse

Same agent. Same four test questions. Same `gpt-4o` judge. The only thing that changes is which tool is watching.

Every screenshot below is from this lab. Every number is measured, not quoted from a docs page.

- **LangSmith**: hosted, at `smith.langchain.com`
- **Langfuse**: self-hosted, six Docker containers on `localhost:3000`, `v4.6.0 OSS`

What we are watching

A small LangGraph agent with three tools:

- `calculator` , always works
- `get_weather` , fake data that **fails 20% of the time on purpose**
- `search_docs` , tiny RAG over five markdown files

The 20% failure is the interesting part. It is what makes the error screens below real instead of staged.

Contents

Document	Covers
01-tracing.md	Turning tracing on, reading a trace, error handling, grouping runs
02-evaluation.md	Golden datasets, LLM-as-judge, measured scores
03-features.md	Dashboards, and what each tool has that the other does not
04-cost.md	Pricing for both, and measured CPU/RAM/disk for self-hosted Langfuse

Running this yourself

```
# plain app, traced by LangSmith automatically
uv run python -m src.agent "what's the weather in Chennai and what's 100/7?"

# same agent, explicitly traced into Langfuse
uv run python -m src.obs_langfuse.traced_run "what's the weather in Chennai and what's 100/7?"

# datasets (run once each)
uv run python -m src.obs_langsmith.dataset
uv run python -m src.obs_langfuse.dataset

# evaluations
uv run python -m src.obs_langsmith.evaluate
uv run python -m src.obs_langfuse.evaluate
```

To force a failure instead of waiting for the 20% dice:

```
PYTHONPATH=. uv run python -c "
import random
random.random = lambda: 0.0
from dotenv import load_dotenv; load_dotenv()
from src.agent.graph import graph
graph.invoke({'messages': [('user', 'what is the weather in Chennai?')]}, config={'run_name': 'for
"
```

Verdict

What you care about	Pick
Tracing working in ten seconds	LangSmith
Data must not leave your network	Langfuse
Seeing <i>where</i> an agent broke	Langfuse
Judge explaining its scores	Langfuse
Nothing to operate	LangSmith
Filtering on your own metadata keys	LangSmith

There is no overall winner. If your data can leave the building, LangSmith gets you further in an afternoon and there is nothing to maintain. If it cannot, Langfuse is a real replacement rather than a downgrade, and its error views are genuinely better. The price is six containers and a port conflict.